

1 Purpose and delivery scope

Distributor Retailer and CSP Lead Portal

Functional and Technical Requirements | Version 1.0 | 18 September 20

Build a centralized web portal to register prospective distributors, retailers and Customer Service Points (CSPs), record their proposed operating locations, verify documents, approve onboarding, issue appointment letters and track territory coverage.

Business outcome

Management must be able to open any state, district or block and see available distributor capacity, current appointments, pending applications and the retailers or CSPs operating there. Each applicant must have one traceable record from first enquiry through activation and later closure.

Required first release

- Lead entry and assignment for Distributor, Retailer and CSP applications.
- Applicant profile, identity details, separate residential and outlet addresses, latitude and longitude.
- Document upload, verification, deficiency requests and review history.
- Due diligence, bank review where applicable, internal approval and activation.
- Template-based appointment letter generation, authorized issue, download, reissue and revocation.
- Territory allocation, vacant and filled block reports, map and geographic drill-down.
- Role-based access, follow-ups, dashboard, Excel reports, audit trail and administration.

Boundaries

This is a lead and onboarding management portal. Banking transactions, AePS, DMT, settlement, commissions, payment collection and core banking integration are outside this release unless separately commissioned. Document upload does not itself perform Aadhaar, PAN or bank verification.

Proposed defaults

The rules in this document are a development baseline. Capacity, required documents, approval roles, expiry periods and letter wording must be configurable. Use company and bank master records rather than hard-coded names. No particular bank approval or exclusive territory right is assumed.

2 Users navigation and permissions

Role	Permitted work
Super Admin	Manage users, role permissions, geographic masters, bank schemes, capacity rules and templates. View all records.
State or District Manager	View assigned territory; assign leads; review progress and coverage. Approve only where separately authorized.
Lead Executive	Create and update assigned leads; upload documents; record calls, visits and follow-ups. Cannot approve own application.
Verification Officer	Check documents and due diligence; accept or return items; record findings. No appointment issue authority by default.
Approver and Letter Issuer	Approve eligible cases; preview and issue approved letters. Permissions may be separated between two users.
Read Only or Auditor	View authorized records, reports and audit events. Sensitive documents and exports require separate permission.
Distributor Portal User	Optional later phase. View only explicitly linked retailers or CSPs; no general applicant or territory database access.

Required screens

Login and password recovery; dashboard; lead list; create or edit lead; lead detail; document review; follow-up calendar; approval queue; partner directory; territory coverage; map; appointment letter register; reports; administration.

Lead detail tabs

Overview, Applicant, Residential Address, Proposed Outlet, Documents, Due Diligence, Follow-ups, Approval, Territory Allocation, Appointment Letters and Activity History.

Access rules

- Enforce role plus assigned state, district or block scope on the server for every screen, API, attachment and export.
- Separate privileges for viewing full identity data, downloading documents, exporting data, issuing letters and changing territory capacity.
- Lead creator cannot be the final approver on the same application. Administrative overrides require a reason and an audit record.
- Users can deactivate accounts and reassign their work without deleting the business history.

3 Applicant and lead data fields

Field gates: C = required to create a lead; S = required before submission for verification; A = required before approval or activation; O = optional. Drafts may be incomplete; the portal must show exactly which fields prevent submission.

Field group	Fields and required behavior
System and routing	Lead ID (automatic, immutable); created date and user; last updated date; applicant ID; application type C; lead owner C; source C; priority O; status automatic.
Applicant identity	Full name C; father name S; date of birth S; applicant photo S; entity type S (individual, proprietorship, company or other); gender O. For entities, capture legal name and authorized representative details.
Contact	Primary mobile C; alternate mobile O; email S. Record mobile/email verification status separately; unverified is not verified by format validation.
Identity references	PAN S under applicable checklist; Aadhaar reference or masked last four digits under the authorized collection policy. Store document type, name match result and verification method. Full Aadhaar is not a default mandatory lead field.
Residential address	Address lines, village or town, post office, state, district, block or urban equivalent, PIN code S. Store separately from the business location.
Application context	Bank or program S for bank-linked applications; proposed services S; parent distributor A for distributor-linked retailer/CSP; direct-company mapping allowed by policy.
Commercial profile	Business/shop name S where applicable; education O; previous banking experience O; existing business O; expected start date O; notes O.
Consent and contact	Capture data collection notice version, purpose, acknowledgment date and method S. Record preferred contact language and communication consent O.

Validation and duplicates

Indian mobile numbers: 10 digits with country code stored separately; email: format check; PIN: six digits; PAN: uppercase format check. Format checks do not prove identity. Normalize mobile and PAN for duplicate screening across leads and partners.

Flag matching mobile or PAN before creation. A match across different types or banks must be reviewed, not blindly rejected. Block a duplicate application for the same applicant, program, type and proposed outlet unless an authorized exception is recorded. Do not expose another applicant's full identity in duplicate warnings.

4 Outlet location and field inspection

Group	Required fields and rules
Operating address	Outlet or shop name; address lines; landmark; village/town; gram panchayat or ward; post office; PIN; rural or urban; state; district; block or mapped urban local body. Required before submission.
Geographic keys	Use stable master IDs for state, district, block and village/ward. Cascading selectors must limit values to the chosen parent. Retain effective dates and inactive master entries for historical records.
Coordinates	Latitude and longitude S for a physical outlet; decimal precision up to six places. Latitude -90 to +90; longitude -180 to +180. Store capture method, timestamp, operator and device accuracy if available.
Capture controls	Capture current GPS with permission; choose a map pin; or enter manually with a reason. Never silently substitute office coordinates or 0,0. Display the selected outlet on a map for confirmation.
Site evidence	Exterior and interior photos; signage photo when applicable; visit date; inspector; GPS captured during visit; findings; inspection outcome. Required fields follow the bank/type checklist.
Infrastructure	Premises ownership or rental; size; electricity; internet; computer; printer/scanner; biometric device; accessibility and security observations. Store Yes/No/Not Applicable with notes.
Mismatch review	Warn when pin and selected administrative area appear inconsistent. If reliable boundary data is available, use point-in-polygon validation. Otherwise require reviewer confirmation; geocoding is not conclusive evidence.

Location lifecycle

A draft location change resets affected location checks. After approval, a material address or coordinate change creates a change request and new verification; it must not silently alter the issued appointment or historical territory assignment.

Master data maintenance

Admin can import and maintain all target states, districts and blocks with source, version and effective date. A block with zero leads must still appear in vacancy reports. Urban applications use an explicit urban unit; never invent a rural block to complete the form.

Multiple outlets

Keep applicant, application and outlet as separate records. One applicant may apply for multiple locations or programs, subject to policy. Each approved outlet receives its own traceable partner mapping.

5 Documents and verification

Document type	Requirement and handling
PAN card	Applicant or entity copy according to the configured checklist; capture review status and name match.
Aadhaar or approved identity proof	Support controlled identity-proof upload and masked Aadhaar where permitted. Restrict access; do not expose the full number in lists or reports.
Applicant photograph	JPG or PNG; linked to the applicant.
Police verification certificate	Configurable requirement by type/program; issuing authority, certificate number, issue date and expiry if applicable.
Company application form	Upload completed form; allow downloading the current blank template. Retain template version used.
Bank application form	Bank/program-specific form and checklist; support multiple signed pages in one PDF.
Due diligence form or letter	Upload signed form and record structured findings, reviewer, date and result.
Appointment letter	System-generated issued PDF plus optional signed acceptance upload; externally issued letter upload must be marked External.
Supporting evidence	Address proof, premises proof, shop photos, education/training certificate, bank approval, cancelled cheque or other program-specific evidence when required.

Upload and review behavior

- Allow PDF, JPG and PNG; proposed limit 10 MB per file, configurable. Validate file signature and MIME type; reject executable, unsafe and password-protected files that cannot be inspected.
- Quarantine new uploads for malware scanning. Only clean files become downloadable or verifiable.
- Store document type, version, original filename, uploader, upload time, issue/expiry dates, verification outcome, reviewer, reviewed time and rejection reason.
- Statuses: Missing, Uploaded, Under Review, Verified, Rejected and Expired. Allow Not Applicable only where the checklist permits and with a reason.
- Replacement creates a new version; do not overwrite evidence used in an approval. Rejected replacements do not inherit Verified status.
- Show a checklist with missing or expired requirements; provide preview and authorized download. Uploaded and Verified are separate states.

Checklist rules

Maintain a versioned checklist by application type and bank/program. Freeze the applicable checklist version at submission. New mandatory requirements for an existing case require an explicit migration or review action with history.

6 Lead workflow and approval gates

Stage	Gate and outcome
Draft or New	Create with minimum fields; assign owner. No reservation, appointment or occupied territory is created.
Contacted or Follow-up	Log call/visit outcome, notes, next action and next due date. Maintain a complete interaction timeline.
Submitted	Validate required profile, location, consent and configured document uploads. Freeze submitted version for review.
Under Verification	Check identity evidence, documents and due diligence. Assign verifier and record findings.
Correction Required	Return specified items with reason. Owner corrects and resubmits; invalidate checks affected by the change.
Pending Bank Approval	Use only for programs requiring external approval. Store submission reference/date and bank decision evidence.
Approved	All mandatory checks passed, bank approval recorded where required, approver decision recorded, valid territory capacity confirmed.
Letter Issued	Authorized user issues an immutable letter for the approved appointment and allocation. Letter alone does not mark the outlet active.
Active	Record activation date, acceptance if required, bank/CSP/partner code where applicable and operational confirmation.
Rejected or Withdrawn	Reason and decision date required. Release reservation. Reopening requires permission and retained history.

Separate lead and partner statuses

After activation, the application remains linked to a partner record. Partner lifecycle: Active, Suspended, Terminated, Expired or Closed. A suspended partner is not reported as actively operating. Retain history rather than moving a closed partner back to a new lead.

Follow-up and escalation

Provide Today, Upcoming and Overdue queues by owner. Notify assigned users in the portal for new assignments, corrections and approvals. Proposed default: flag leads with no update for seven days; Admin can change this threshold. External SMS, email and WhatsApp sending are optional integrations.

Concurrency and edits

Use version checks so one user cannot unknowingly overwrite another user's review. Require reasons for reassignment, rejection, reopening and post-approval edits. Changes to applicant identity, outlet or program reset dependent approvals and may require a revised appointment.

7 Appointment letter generation and issue

Template management

Provide separate editable templates for Distributor, Retailer and CSP appointments, selectable by company and bank/program. An authorized administrator manages letterhead, clauses, signatory and validity rules. Templates have Draft and Approved versions; only approved versions may be issued.

Merge field group	Information populated from approved records
Reference	Unique appointment number; issue date; application and partner IDs; template version.
Appointee	Approved full/legal name; father name if included in the template; business name; correspondence address; contact details.
Appointment	Distributor/Retailer/CSP role; bank/program; approved services; effective date; expiry date or no-expiry policy.
Territory and outlet	Approved outlet address; state, district, block and additional allocated blocks; parent distributor if applicable.
Authority and conditions	Issuing company; authorized signatory name/designation; approved terms; acceptance deadline and signature section.
Verification	Optional QR code or verification URL showing only letter number, role, issuing company, validity and current status.

Controlled issue process

Select approved application → select approved template → preview as DRAFT → validate mandatory merge fields and allocation → authorized issue → store PDF and issue event → download/print. Allocate the reference number transactionally; repeat clicks or retries must not create duplicate letters.

- Proposed reference format: COMPANY/TYPE/YEAR/SEQUENCE. Configure the actual prefix and reset policy before launch.
- Freeze a snapshot of all merged data and the issued PDF. Later profile or template edits must not rewrite an existing letter.
- A signature image may be applied only by an authorized issuer with controlled access. Cryptographic digital signing requires a separately configured provider.
- Reissue creates a new version or reference linked to the earlier letter and marks the earlier letter Superseded. Revoke requires reason, date and authorized action.
- Maintain Draft, Issued, Superseded, Revoked and Expired states; record download and acceptance history. Expiry must respect any explicitly approved no-expiry policy.

Authority boundary

The company appointment must not claim to be a bank-issued authorization. For a bank-linked CSP, require recorded bank approval before issuing wording that confirms bank authorization. Business owners supply and approve final clauses and signatory authority.

8 Territory allocation and block availability

Capacity model

Track capacity by company, bank/program, partner type and geographic unit. Proposed initial distributor capacity is one slot per block per program, subject to Admin confirmation. Retailer and CSP capacity is separate and may be configured as unlimited; neither consumes a distributor slot.

State	Definition
Not Open	Unit is outside the configured rollout or has zero allowed capacity. Keep it separate from Vacant.
Vacant	Unit is open; zero occupied slots; zero reserved slots; available capacity remains.
Reserved	Zero occupied slots but at least one unexpired reservation. Show whether any additional slots remain available.
Partially Filled	At least one occupied slot but occupied slots are below configured capacity. Display occupied, reserved and available separately.
Filled	Occupied slots equal capacity. Do not infer this from the number of leads.
Available slots	Maximum of capacity minus occupied minus reserved, or zero. Pending unreserved leads do not reduce availability.

Allocation and reservation rules

- An active, issued appointment allocation occupies capacity from its effective date. Operationally active outlets are counted separately after activation.
- Reservations belong to one application and expire automatically. Proposed reservation period: seven days, configurable; extension needs a reason and authorized user.
- Prevent overlapping appointments above capacity with server-side checks and transactional locking. Never rely only on a screen warning.
- One distributor may hold several blocks if permitted. Keep one allocation row per block with start/end dates, status and appointment reference.
- Proposed suspension rule: retain the allocation as occupied but display On Hold. Termination, expiry or closure releases it unless an explicit approved hold applies.
- Transfers close the previous allocation and open the new one with effective dates. Do not delete the former distributor from history.
- For unlimited capacity, report active counts and mark capacity Unlimited; do not label the unit Filled.

Coverage screen

Provide State → District → Block drill-down and table/map views. Each row shows configured capacity, occupied/reserved/available slots, distributor names and IDs, active retailers, active CSPs, pending leads and latest update. Filter by program, type and current or historical date.

Example: a block with capacity 1, three unapproved distributor leads and no reservation remains Vacant. One effective distributor appointment makes it Filled even if outlet activation is still pending; active operating count remains zero until activation.

9 Dashboard reports and bulk operations

Report	Required contents
Management dashboard	Leads by type/stage; pending verification; pending approval; overdue follow-ups; issued letters; active partners; coverage and expiry alerts.
Lead register	Lead ID, applicant name, type, program, mobile masked by permission, location, owner, stage, created date and next follow-up.
Vacant and occupied blocks	All configured units including zero-lead blocks; capacity, occupancy, reservations, available slots and allocated distributor details.
Network directory	State/district/block-wise distributors and their linked retailers/CSPs; outlet address, partner code, status and activation date.
Documents and exceptions	Missing, rejected and expiring documents; pending corrections; duplicate-review cases; location mismatch cases.
Appointment register	Letter reference, appointee, type, issuer, issue/effective/expiry dates, status and superseded reference.
Performance and aging	Owner-wise leads, submissions, approvals and activations; stage aging; daily/weekly/monthly trends. Show date basis and denominator.
Audit report	Who changed what and when; decision reasons; territory changes; letter issue/revoke; sensitive document accesses and exports.

Filters and exports

Support date range, state, district, block, type, bank/program, distributor, owner and status. Search by lead/partner ID, name, mobile and permitted reference fields. Provide paginated results, Excel/CSV exports and printable reports; exported totals must match the same filters and permissions.

Use distinct partner/outlet IDs for totals. A distributor covering multiple blocks must not inflate the unique distributor count. Show coverage as filled eligible blocks divided by open configured blocks for the selected program and capacity policy; show retailer/CSP presence separately.

Bulk lead import

Provide an Excel template with required fields and master codes. Validate in a preview stage, flag duplicates and invalid geography, show row-level errors, then commit only approved valid rows. Retain batch ID and import summary; retrying a batch must not duplicate accepted records. Bulk import cannot create verified or approved applications.

Alerts

Display overdue follow-ups, reservation expiry, document expiry and appointment expiry in the portal. Proposed reminders at 30, 7 and 1 days before applicable expiry, configurable. Queue optional external notifications with consent checks, delivery status and retries; failure must not undo the underlying approval.

10 Security privacy and operating quality

Sensitive information controls

- Use HTTPS, secure authentication, strong password controls and MFA for administrators, approvers and letter issuers. Apply session timeout and revoke sessions for disabled users.
- Encrypt sensitive fields and documents at rest. Keep secrets and encryption keys outside application code; separate production and test environments.
- Store documents privately; use short-lived authorized download links. Do not expose public bucket paths or place Aadhaar/PAN values in URLs, filenames, logs or analytics.
- Display masked identity numbers by default. Full identity access, where authorized, requires a specific permission and an audit event.
- Before enabling full Aadhaar collection or authentication, confirm the permitted purpose, bank policy, consent and approved service-provider route. Support masked or alternative identity evidence according to the accepted checklist.
- Maintain a retention schedule approved by the business owner, including rejected leads, old document versions and backups. Deletion or anonymization must be controlled and preserve required audit references.

Availability and recovery targets

Proposed targets for quotation: 100 concurrent internal users, 100,000 lead records and one million document metadata entries. Typical paginated screens should load within three seconds at the 95th percentile under the agreed test environment; large exports run as background jobs.

Proposed recovery targets: daily encrypted backups with recovery point within 24 hours and restoration within 8 hours. Conduct a restoration test before handover. Supplier must state hosting capacity, uptime target, monitoring, support coverage and recurring costs.

Usability and reliability

Responsive web interface for desktop and mobile browsers; Unicode names and addresses; save draft; clear field-level errors; confirmation for destructive or irreversible actions. GPS permission denial must allow manual entry with a reason. Offline mode is outside the initial scope.

Store timestamps consistently and display business dates in Asia/Kolkata. Use a configurable business timezone for reference-year and expiry calculations. Background jobs must retry safely, show failure states and alert administrators.

Audit integrity

Audit events are append-only for ordinary users and administrators; corrections are additional events. Capture actor, action, entity, time, permitted before/after differences and reason. Never copy raw identity numbers or document contents into the audit log.

11 Data model and integration contract

Entity	Relationships and key constraints
Applicant and Contact	Applicant owns identity/profile and contacts. One applicant can have several applications. Sensitive references are separate from ordinary searchable profile data.
Lead Application	Links applicant, type, program, owner, workflow state and submitted version. One application targets one outlet and appointment scope.
Outlet and Geography	Outlet holds location/GPS and evidence. Geography masters use stable IDs and parent references, not only free-text names.
Partner and Parent Mapping	Approved partner links applicant, type, program and outlet. Retailer/CSP links to a distributor or direct company; effective-date history retained.
Capacity and Allocation	Capacity keyed by program/type/unit; allocation links appointment, partner and unit with effective dates. Reservation has owner and expiry.
Document and Review	Document versions belong to application/applicant/outlet as appropriate. Review refers to exact version and checklist requirement.
Approval and Due Diligence	Decision actor, evidence, checklist version, outcome and reason. Preserve independent internal and bank decisions.
Letter and Template	Versioned template; immutable issued data/PDF; reference, status, signatory, issue and validity dates.
User and Activity	Roles plus scope, follow-ups, consent records, audit events, notification jobs and import batches.

Application architecture

Use a responsive frontend, authenticated backend API, relational database, private object storage and background workers for document scanning, letters, reminders and exports. Supplier may propose the technology stack but must document maintainability, licenses and deployment requirements.

API behavior

Document lead create/update/submit, document upload/review, workflow transition, allocation/reservation, letter preview/issue/revoke and coverage/report APIs. Apply server-side validation and permission checks. Use record versions for edits and idempotency keys for issue/import/activation operations.

Integrations

First release may use manual evidence for bank approval and document verification. Maps/geocoding, SMS/email, WhatsApp, PAN verification, e-sign and bank APIs need separately approved providers, credentials, costs and data contracts. Keep provider configuration replaceable and record the source of each verification result.

12 Acceptance tests for developer handover

The following tests must pass in staging using fictional or masked test data. Business users must be able to demonstrate each result before production acceptance.

ID	Test and expected result
UAT 01	Create each lead type; save an incomplete draft; block submission with specific missing-field messages. All profile and outlet details persist after login.
UAT 02	Change state or district: incompatible child selections are cleared. A valid outlet pin is saved; invalid coordinates are rejected; denied GPS permission offers manual capture.
UAT 03	Upload each required document; reject unsafe/oversized files. Replace a verified document: new version requires review and earlier evidence remains accessible to authorized users.
UAT 04	Attempt to open another territory's lead or private attachment through a direct URL/API: deny access. Verify that exports obey the same scope and masking.
UAT 05	Create matching mobile/PAN applications: show a scoped duplicate alert. Block exact duplicate case unless an authorized exception is recorded.
UAT 06	Attempt approval with missing verification or required bank decision: block it. Creator cannot approve own case. Corrections preserve submission and review history.
UAT 07	Create leads in an empty block: block stays Vacant. Add a reservation: available capacity falls. Expire/cancel it: availability returns.
UAT 08	Two users attempt the final distributor slot simultaneously: only one allocation succeeds. Retailer/CSP activation does not consume that distributor slot.
UAT 09	Issue a letter twice via retry: produce one issued reference. Profile/template changes do not change the existing PDF. Reissue links versions and revocation updates status.
UAT 10	Letter issuance alone does not increase active operating CSPs. Activation needs configured evidence and codes. Suspension shows On Hold; termination releases capacity per policy.
UAT 11	Coverage includes blocks with no leads; multi-block distributors are counted once in unique partner totals. Historical report reproduces prior allocations.
UAT 12	Import mixed valid/invalid rows: preview errors, commit valid rows only, retry without duplication. Export totals match screen filters; sensitive fields remain controlled.
UAT 13	Restore a backup in staging; confirm linked files, records and letter PDFs. Test role revocation, audit history and agreed performance targets.

13 Delivery plan and configuration signoff

Suggested implementation sequence

Milestone	Reviewable output
1 Design and setup	Screen designs, field dictionary, role matrix, data model, master-data import and checklist configuration.
2 Lead and document module	Working forms, geographic capture, assignments, follow-ups, document upload and review.
3 Approval and territory module	Workflow gates, bank-decision evidence, capacity/reservations, allocation history and coverage screens.
4 Letters and reports	Template approval, controlled PDF issue, letter register, dashboards, import/export and reminders.
5 Validation and launch	UAT evidence, security review, restore/performance checks, user training and production deployment.

Developer handover package

- Working application and source code repository under company control, with build and deployment instructions.
- Database schema, migrations, data dictionary, API documentation and integration configuration guide.
- Admin and user manuals; training session; UAT checklist with results and known limitations.
- Company-controlled hosting and service accounts; private document storage, backups, monitoring and restore procedure.
- Inventory of dependencies, licenses, domains and third-party services; agreed warranty, support response times and maintenance terms.

Configuration decisions before production

Confirm issuing company and branding; supported banks/programs; signatories and final letter wording; state/district/block master source; initial rollout area; per-type capacity and exclusivity policy; reservation period; approval roles; required documents; retention rules; parent mapping; bank code/activation requirements; language options and integration providers.

The supplier quotation must separate one-time development, hosting/storage, map usage, verification APIs, messaging, digital signing and ongoing support. No external integration should be assumed free or available without an approved account.

Scope acceptance

Business owner: _____ Date: _____

Developer / supplier: _____ Date: _____

Approved changes or exclusions: _____
